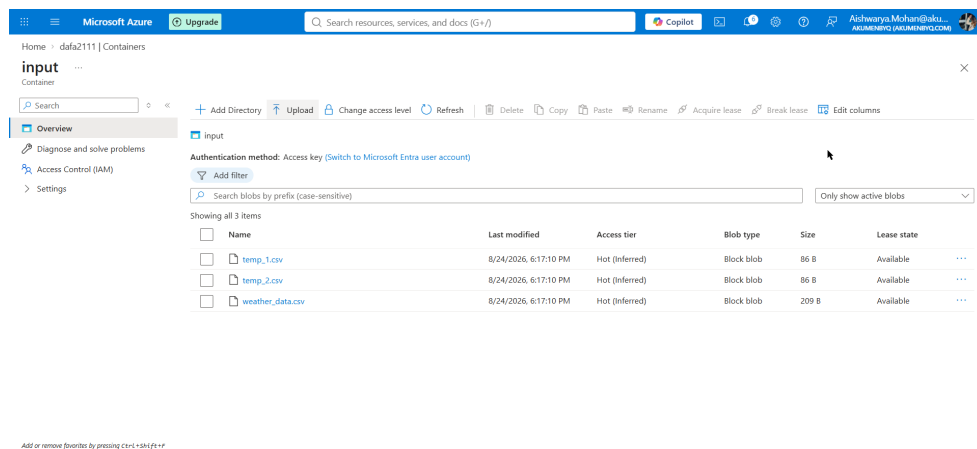
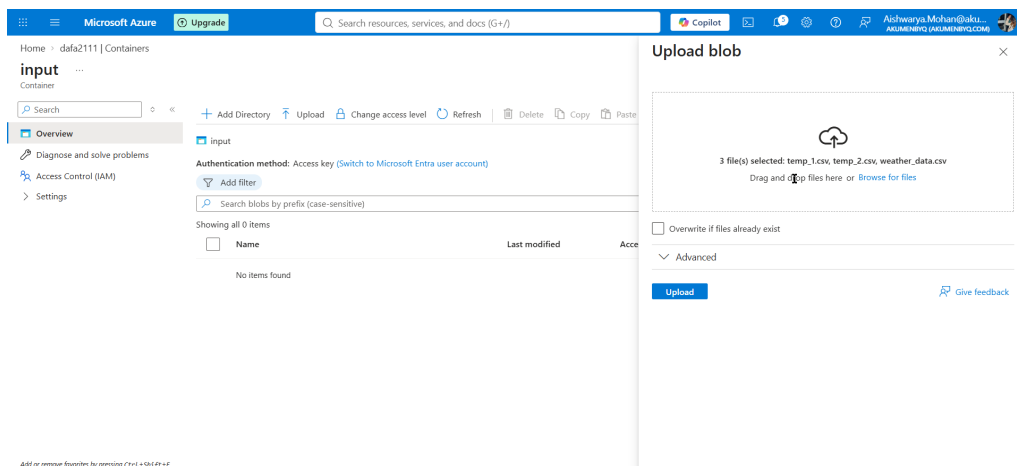
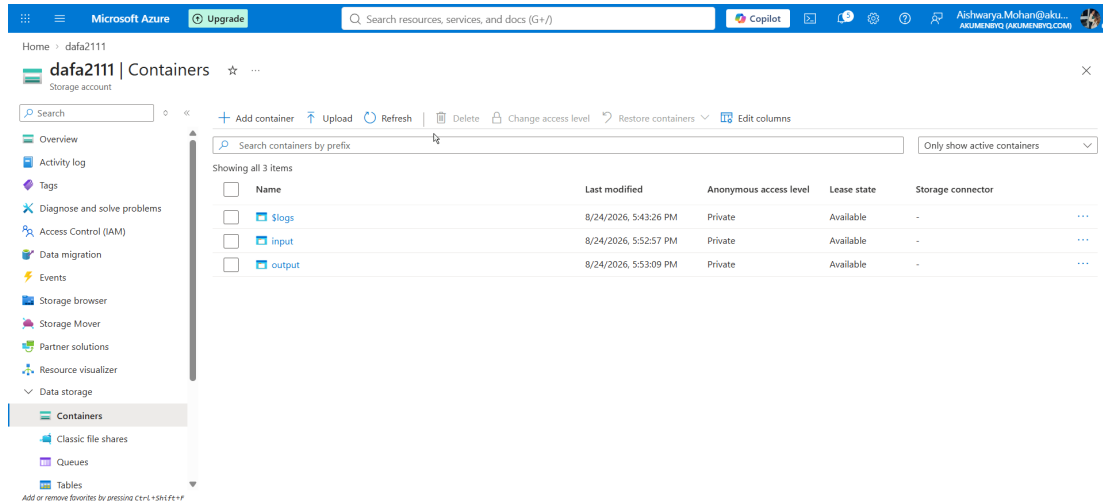


Azure Data Factory – Control Flow

Get Metadata → ForEach → If Condition

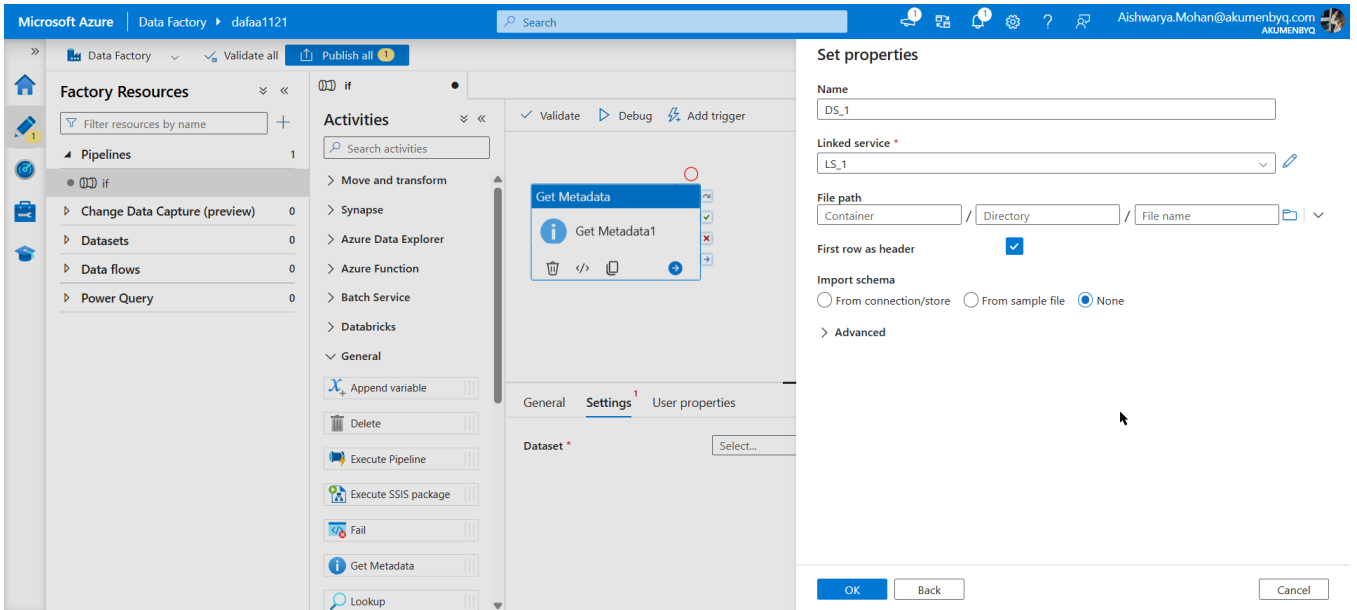
① Storage Setup

Create containers **input** & **output** → Upload 3 files to input

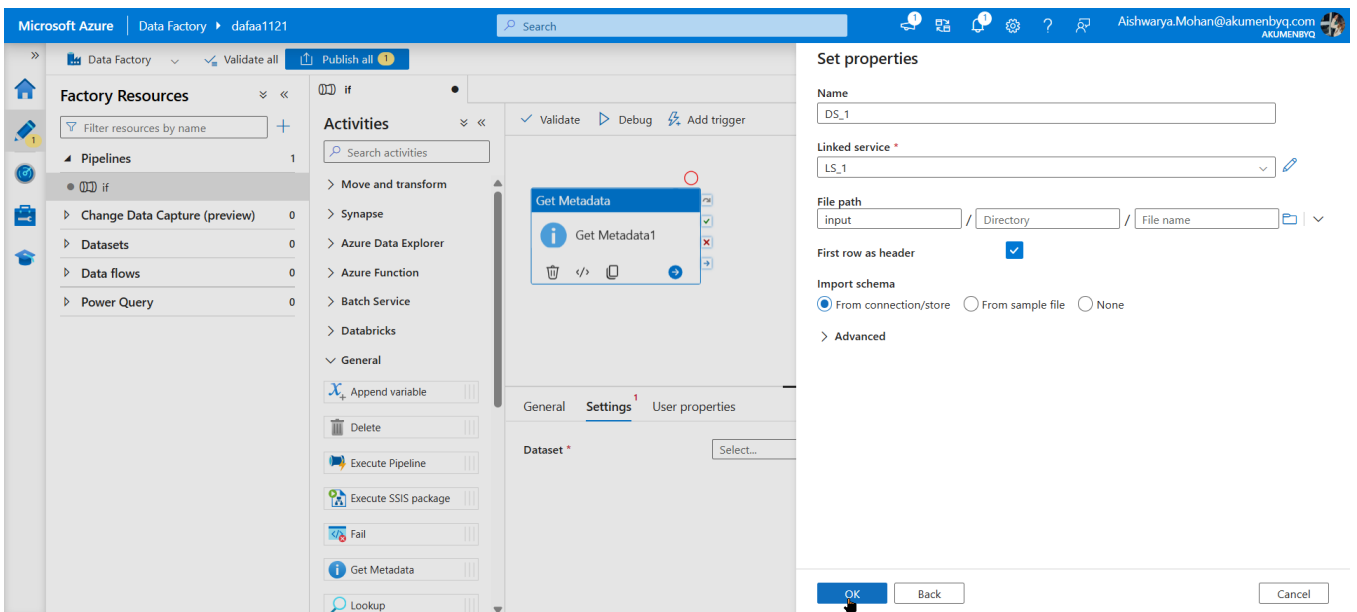


② Get Metadata Activity

Dataset DS_1 → Container = input | Field list = Child items



Create dataset DS_1 linked to input container



File path set to input | Import schema = From connection/store

③ ForEach + If Condition

Items = @activity('Get Metadata1').output.childItems
Expression = @contains(item().name,'temp')

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane is visible, showing a hierarchy of resources: Pipelines (1), if (1), Change Data Capture (preview) (0), Datasets (1), Data flows (0), and Power Query (0). The 'Activities' pane on the right shows a list of activities: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. The 'Get Metadata' activity is selected, and its configuration is shown in the 'Settings' tab. The 'Dataset' is set to 'DS_1', and the 'Field list' is set to 'Child items'. The 'Properties' pane on the right shows the 'General' tab with the 'Name' set to 'if'.

Get Metadata1 connected → Field list = Child items

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane is visible, showing a hierarchy of resources: Pipelines (1), if (1), Change Data Capture (preview) (0), Datasets (1), Data flows (0), and Power Query (0). The 'Activities' pane on the right shows a list of activities: Move and transform, Synapse, Azure Data Explorer, Azure Function, Batch Service, Databricks, and General. The 'ForEach' activity is selected, and its configuration is shown in the 'Settings' tab. The 'Concurrency' is set to 1, and the 'Elapsed time metric' is set to '0.00:10:00'. The 'Properties' pane on the right shows the 'General' tab with the 'Name' set to 'if'.

ForEach1 added after Get Metadata1

④ ForEach Settings + Expression

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Factory Resources

- Pipelines
- if
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

Activities

- for
- Move and transform
 - Copy data
 - Data flow
- Iteration & conditionals
 - ForEach

General Settings Activities (0) User properties

Sequential ☐

Batch count

Items *

This property should be parameterized.

Add dynamic content (Alt+Shift+D)

Properties

General Related

Name *

Description

Annotations

+ New

Cancel

Items property must be parameterized

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Factory Resources

- Pipelines
- if
- Change Data Capture (preview)
- Datasets
- Data flows
- Power Query

Activities

- for
- Move and transform
 - Copy data
 - Data flow
- Iteration & conditionals
 - ForEach

General Settings Activities (0) User properties

Sequential ☐

Batch count

Items *

This property should be parameterized.

Add dynamic content (Alt+Shift+D)

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

Clear contents

Activity outputs Parameters System variables Functions Variables

Get Metadata1
Get Metadata1 activity output

Get Metadata1 childItems
List of subfolders and files in the given folder

Get Metadata1 exists
Whether a file, folder, or table exists

Get Metadata1 itemName
Name of the file or folder

OK Cancel

Expression builder → select childItems

⑤ Add If Condition inside ForEach

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' with a count of 1. The main canvas displays a pipeline with a 'Get Metadata' activity followed by a 'ForEach' activity named 'ForEach1'. The 'Add trigger' dialog is open, showing the 'If' condition type. The 'Items' field is set to '@activity("Get Metadata1").output.chi...'. The 'Properties' pane on the right shows the 'General' tab with the name 'if' and a description field.

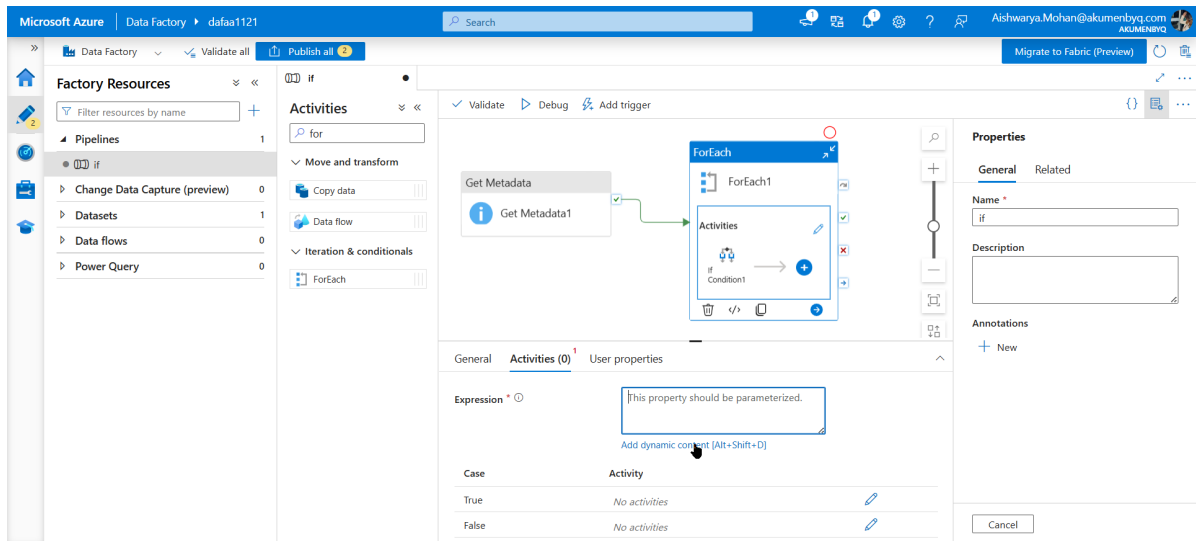
Search & drag If Condition inside ForEach activities

The screenshot shows the Microsoft Azure Data Factory interface. The 'If Condition' activity has been added inside the 'ForEach' activity. The 'General' tab of the 'If Condition' activity is selected, showing the name 'If Condition1' and the description field. The 'Activity state' is set to 'Activated'.

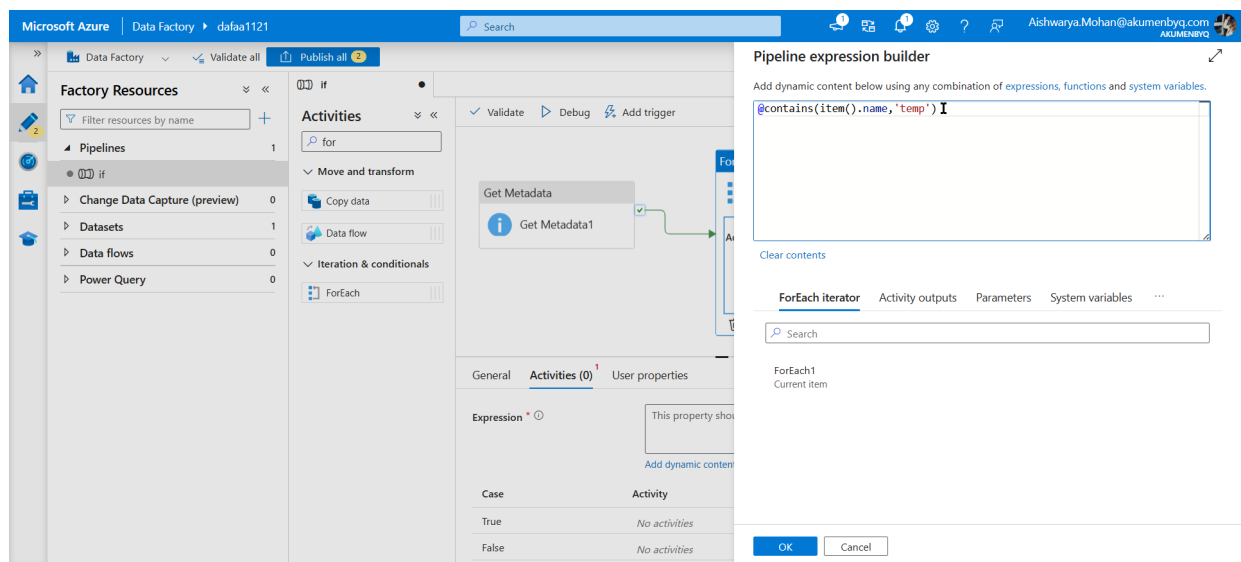
If Condition1 appears inside ForEach

⑥ True / False Branches with Set Variable

True → Set variable temp = True
False → Set variable weather = False

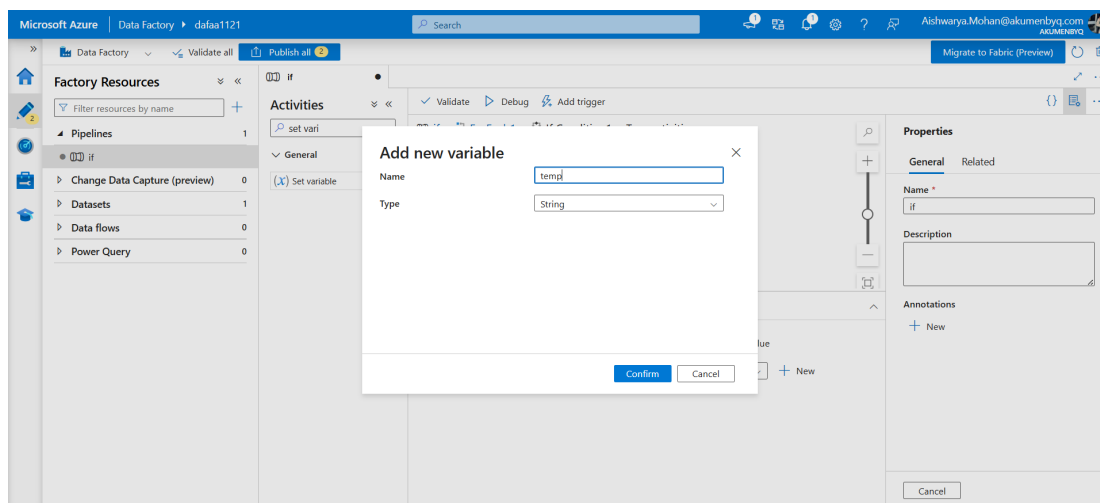


Expression property → Add dynamic content

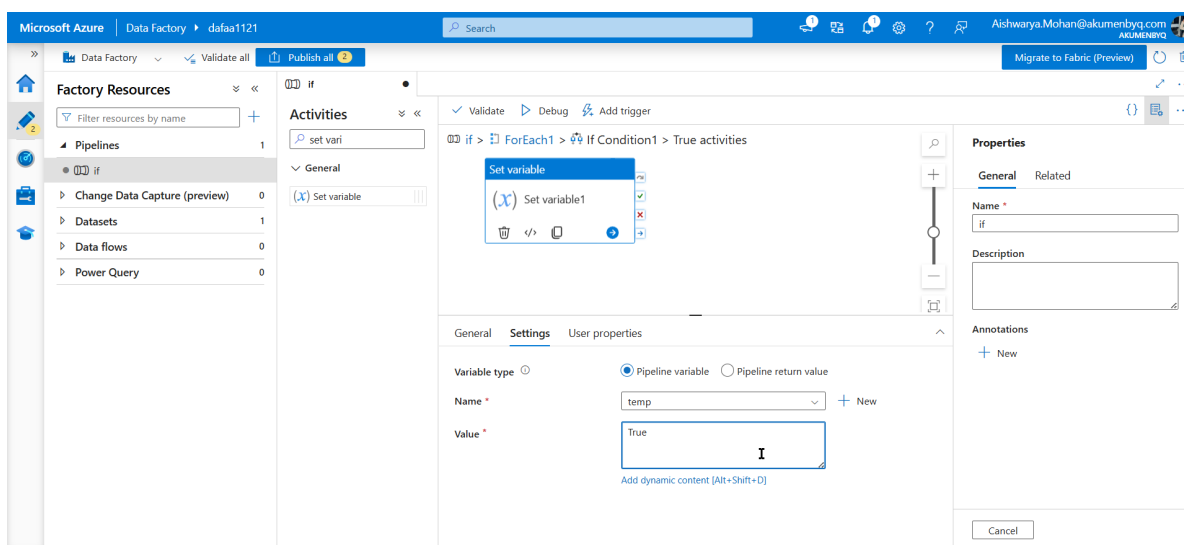


@contains(item().name, 'temp')

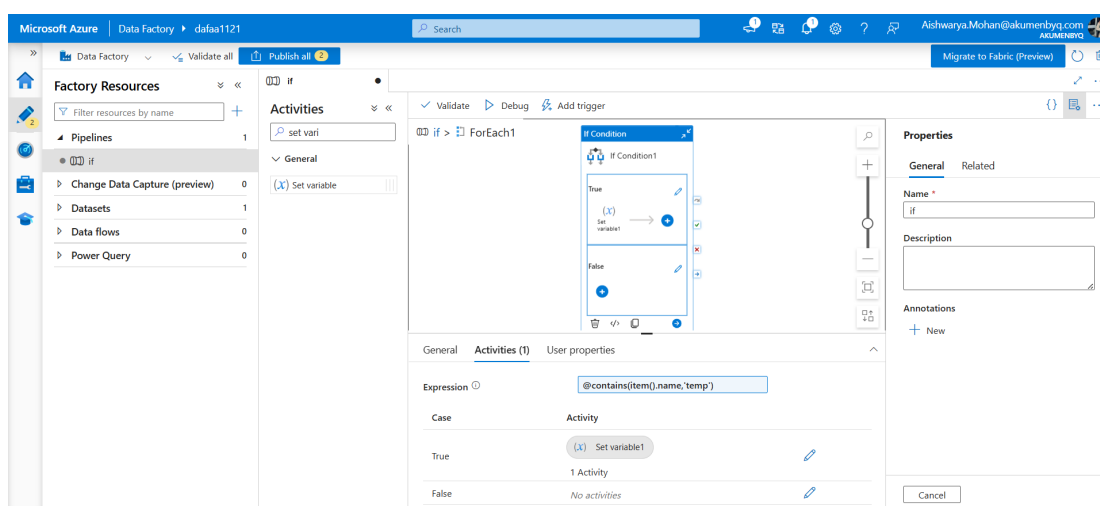
⑦ Create Variables & Set Values



Add new variable: temp (String)



True branch → Set variable1 → Value = True



If Condition shows True: Set variable1 | False: empty

⑧ False Branch Set Variable

The screenshot shows the Microsoft Azure Data Factory editor interface. On the left, the 'Factory Resources' pane lists 'Pipelines' with 1 item, 'Datasets' with 1 item, 'Data flows' with 0 items, and 'Power Query' with 0 items. The main canvas displays an 'if' activity with two branches. The 'False' branch is selected, and the 'Set variable' activity is configured with the following properties:

- Variable type:** Pipeline variable
- Name:** weather
- Value:** False

The 'Properties' pane on the right shows the 'General' tab with the Name 'if' and a description field.

False branch → Set variable2 → weather = False

The screenshot shows the Microsoft Azure Data Factory editor interface. On the left, the 'Factory Resources' pane lists 'Pipelines' with 1 item, 'Datasets' with 1 item, 'Data flows' with 0 items, and 'Power Query' with 0 items. The main canvas displays an 'if' activity with two branches. The 'True' branch is selected, and the 'Set variable' activity is configured with the following properties:

- Expression:** @contains(item().name,'temp')
- Case:** Activity
- True:** 1 Activity
- False:** 1 Activity

The 'Properties' pane on the right shows the 'General' tab with the Name 'if' and a description field.

Both True & False branches now have Set variable

⑨ Debug Result (Set Variable version)

All activities Succeeded

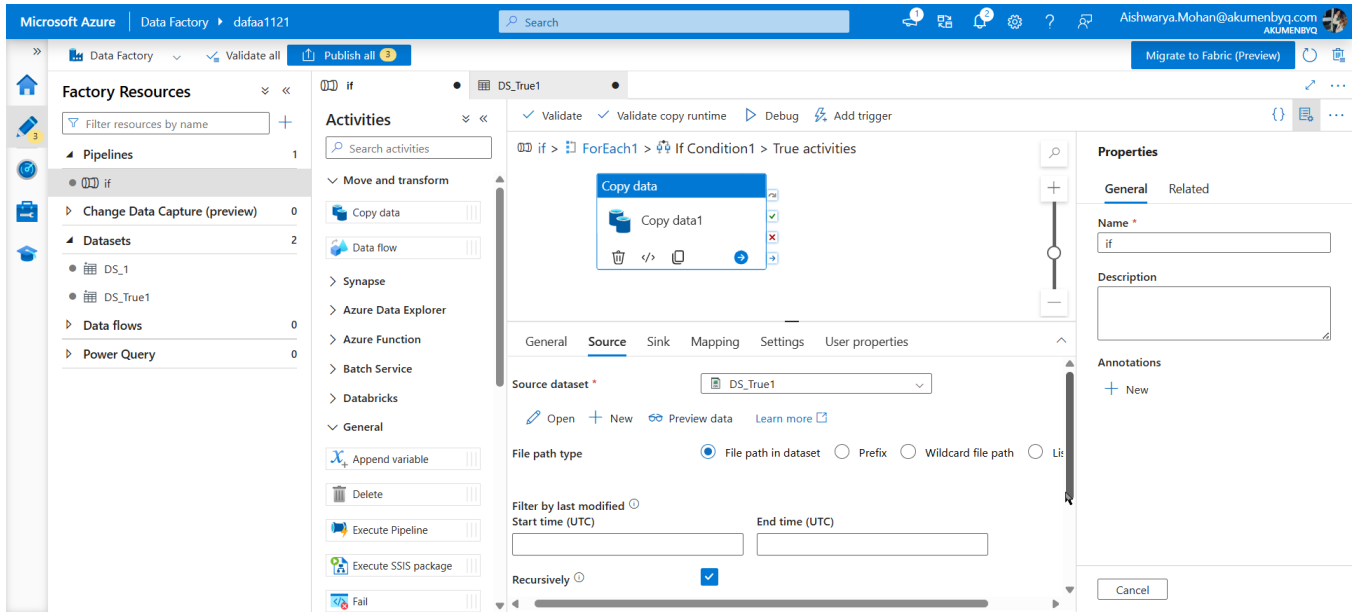
The screenshot displays the Microsoft Azure Data Factory interface for a pipeline named 'if'. The pipeline is in a 'Succeeded' state, and the 'Debug' tab is active. The pipeline diagram shows a sequence of activities: 'Get Metadata' (Get Metadata1), 'Set variable' (Set variable), 'If Condition' (If Condition1), and 'ForEach' (ForEach1). The 'Output' tab is selected, showing a table of activity results.

Activity name	Activity status	Activity name	Run start	Duration
Set variable2	✓ Succeeded	Set variable	8/24/2026, 6:49:36 PM	Les
Set variable1	✓ Succeeded	Set variable	8/24/2026, 6:49:36 PM	Les
Set variable1	✓ Succeeded	Set variable	8/24/2026, 6:49:36 PM	Les
If Condition1	✓ Succeeded	If Condition	8/24/2026, 6:49:36 PM	2s
If Condition1	✓ Succeeded	If Condition	8/24/2026, 6:49:36 PM	2s
If Condition1	✓ Succeeded	If Condition	8/24/2026, 6:49:36 PM	2s
ForEach1	✓ Succeeded	ForEach	8/24/2026, 6:49:35 PM	4s
Get Metadata1	✓ Succeeded	Get Metadata	8/24/2026, 6:49:16 PM	18s

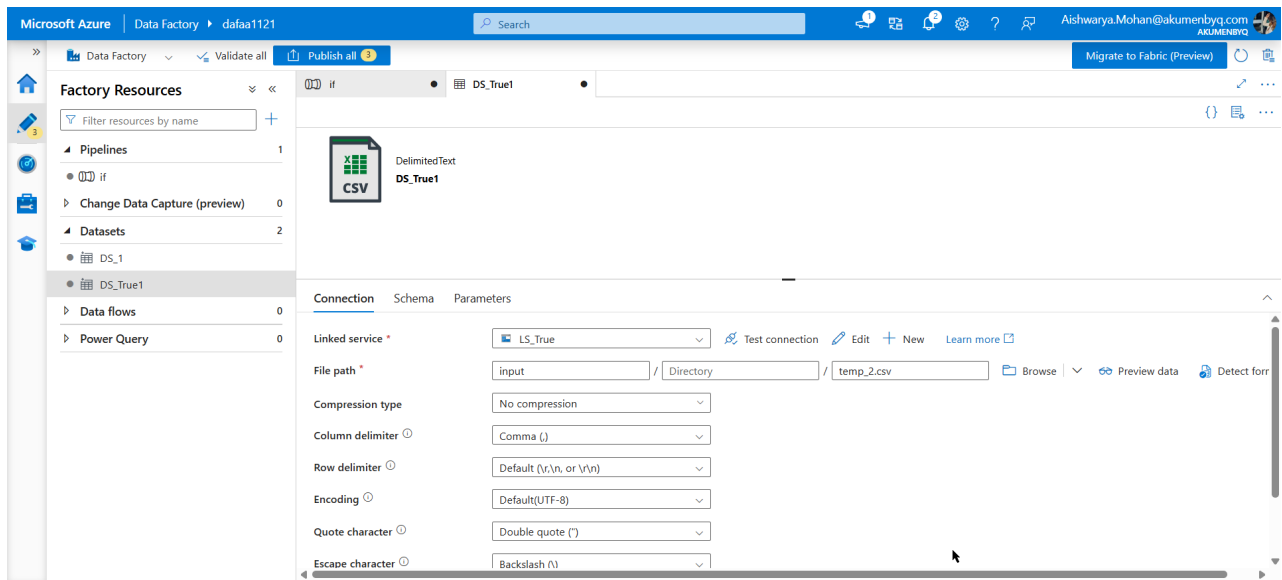
Pipeline run output – Get Metadata, ForEach, If Condition, Set variables all green

10 Replace Set Variable with Copy Data

True → Copy data1 | False → Copy data2



Copy data1 inside True activities – Source = DS_True1



Dataset DS_True1 points to input container

Final Pipeline with Copy Activities

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Factory Resources

- Pipelines: 1
 - if
- Change Data Capture (preview): 0
- Datasets: 4
 - DS_1
 - DS_False
 - DS_False1
 - DS_True1
- Data flows: 0
- Power Query: 0

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail

Properties

General

Name: if

Description:

Annotations: + New

Activity state: Validated

Pipeline validation output

Your pipeline has been validated.
No errors were found.

If Condition1 → True: Copy data1 | False: Copy data2

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Factory Resources

- Pipelines: 1
 - if
- Change Data Capture (preview): 0
- Datasets: 4
 - DS_1
 - DS_False
 - DS_False1
 - DS_True1
- Data flows: 0
- Power Query: 0

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
 - Append variable
 - Delete
 - Execute Pipeline
 - Execute SSIS package
 - Fail

Properties

General

Name: if

Description:

Annotations: + New

Activity state: Validated

Pipeline validation output

Your pipeline has been validated.
No errors were found.

Pipeline validated successfully – No errors found

■ Pipeline Run & Output Container

Files copied to output container

The screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists 'Pipelines' and 'Datasets'. The 'Pipelines' section shows a pipeline named 'if'. The 'Datasets' section lists 'DS_1', 'DS_False', 'DS_False1', and 'DS_True1'. The main canvas displays the pipeline logic: an 'If' condition with two branches. The 'True' branch contains a 'Copy data' activity named 'Copy data1'. The 'False' branch contains a 'Copy data' activity named 'Copy data2'. The 'Properties' pane on the right shows the 'General' tab for the 'if' pipeline, with fields for 'Name' and 'Description'. The 'Annotations' section shows a '+ New' button.

Debug run – All Copy data & If Condition activities Succeeded

The screenshot shows the Microsoft Azure Storage Explorer interface. The 'output' container is selected, and the 'Overview' tab is active. The 'Authentication method' is set to 'Access key (Switch to Microsoft Entra user account)'. The 'Add filter' button is visible. The 'Search blobs by prefix (case-sensitive)' field is empty. The 'Only show active blobs' dropdown is set to 'Only show active blobs'. The table below shows the contents of the container:

Name	Last modified	Access tier	Blob type	Size	Lease state
Pokemon.csv	8/24/2026, 7:10:04 PM	Hot (Inferred)	Block blob	4.6 KiB	Available
temp_2.csv	8/24/2026, 7:18:41 PM	Hot (Inferred)	Block blob	86 B	Available

Add or remove favorites by pressing Ctrl+Shift+F

output container now contains the copied files

The screenshot shows the Microsoft Azure Storage Explorer interface. The 'input' container is selected, and the 'Overview' tab is active. The 'Authentication method' is set to 'Access key (Switch to Microsoft Entra user account)'. The 'Add filter' button is visible. The 'Search blobs by prefix (case-sensitive)' field is empty. The 'Only show active blobs' dropdown is set to 'Only show active blobs'. The table below shows the contents of the container:

Name	Last modified	Access tier	Blob type	Size	Lease state
Pokemon.csv	8/24/2026, 7:18:40 PM	Hot (Inferred)	Block blob	4.6 KiB	Available
temp_1.csv	8/24/2026, 6:17:10 PM	Hot (Inferred)	Block blob	86 B	Available
temp_2.csv	8/24/2026, 7:12:45 PM	Hot (Inferred)	Block blob	86 B	Available
weather_data.csv	8/24/2026, 6:17:10 PM	Hot (Inferred)	Block blob	209 B	Available

Add or remove favorites by pressing Ctrl+Shift+F

input container still has original files + Pokemon.csv

Logic: Get Metadata → ForEach → If contains('temp') → True: Copy | False: Copy